Cadet: _____

Section: _____

WPR 1 Practice Examination

Descriptive and Predictive Modeling — Blocks 1 and 2

This is unofficial review material prepared by the course CIC for the WPR 1 study session. It is **not** the WPR and is not drawn from any secured assessment. Coverage follows the lesson objectives for Lessons 1–21 and the conventions of Homeworks 1 and 2. Per the course director, the emphasis is **conceptual understanding and precise mathematical interpretation**, not Excel mechanics.

General Instructions: Read carefully.

1. You have **105 minutes** to complete this practice examination. It is worth **170 points**. (The WPR itself is 150 points in 95 minutes; this runs slightly long on purpose.)
2. Authorized items: your issued calculator. Work it as you would the WPR — individually, closed notes, no electronics beyond the calculator.
3. **Show your work.** When in doubt, explain too much rather than too little. When asked to verify something, show the arithmetic.
4. Round statistics to **two decimal places** unless told otherwise, carrying unrounded values through intermediate steps. Every interpretation must be written **in context**, naming the variable and its units — an answer with no units is incomplete.
5. Answer in the spaces provided; if you run out of room, use the continuation sheet at the back and label it with the problem number. Suggested pacing is printed at the head of each part — if you fall behind, move on and return.

Point allocation

Part	Topic	Points	Suggested time
I	Modeling Framework & Data	24	14 min
II	Descriptive Modeling	56	32 min
III	Linear Regression & Residuals	50	32 min
IV	K-Nearest Neighbors (classification)	32	22 min
V	Synthesis	8	5 min
Total		**170**	**105 min**

I ACKNOWLEDGE THAT I HAVE READ THE ABOVE INSTRUCTIONS AND WILL WORK THIS PRACTICE EXAMINATION UNDER THE SAME CONDITIONS AS A GRADED EVENT.

SIGNATURE: _____ DATE: _____

Situation

You are the operations officer for Company B, 2nd Regiment. The company runs a 12-mile ruck march at the end of every term. The commander wants to use last term's data to (1) describe how the company actually performed and (2) predict how individual cadets will perform this term, so that at-risk cadets can be placed on a remedial training plan before the march.

All ruck times are recorded in **minutes**. All cadets carried a 35-pound rucksack over the same route. The Army standard for the 12-mile foot march is **180 minutes**; a cadet who finishes in 180 minutes or less receives a **GO**.

FORMULA SHEET

Sample mean

$$\bar{x} = \frac{\Sigma x_i}{n}$$

Sample standard deviation

$$s = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n - 1}}$$

Interquartile range

$$\text{IQR} = Q_3 - Q_1 \qquad \text{Range} = \max - \min$$

z-score (standardizing)

$$z_i = \frac{x_i - \bar{x}}{s}$$

Regression line and residual

$$\hat{y} = b_0 + b_1 x \qquad e_i = y_i - \hat{y}_i$$

Mean absolute error

$$\text{MAE} = \frac{\Sigma |y_i - \hat{y}_i|}{n}$$

Distance in standardized space

$$d = \sqrt{(z_{1,i} - z_1^*)^2 + (z_{2,i} - z_2^*)^2}$$

KNN starting value of k

$k \approx \sqrt{n_{\text{train}}}$, rounded to the nearest **odd** integer

Notes to the cadet

- Read every interpretation prompt twice. Most lost points on this material are interpretation points, not arithmetic points.
- Ruck time and run time are variables where **larger is worse**. Watch the direction of your z-scores and your slope.
- When a question gives you a model, use the model it gives you — do not re-derive it from the data table.
- Several questions ask you to carry a value forward from an earlier question. If you could not get the earlier value, state what you are assuming and continue; you will receive follow-on credit.

Part I — Modeling Framework and Data

24 POINTS · APPROXIMATELY 14 MINUTES · LESSONS 1–3, 8

1. Descriptive or predictive? Label each question below. Then, for question (b) only, state in one sentence what makes it that type. *4 pts*

a. What was the average 12-mile ruck time in Company B last term? _____

b. What ruck time should we expect from a cadet who trains 25 miles per week? _____

c. Which of the four platoons had the largest spread in ruck times? _____

2. Population, sample, and data dictionary. *7 pts*

(a) The commander's question is about *this term's* Company B cadets, but your data is *last term's* 64 records. Identify the **population** of interest and state whether your data is the full population or a sample. How do you know?

(b) Complete the data dictionary. For each column give the variable type (**categorical**, or **numerical** and then **discrete** or **continuous**) and the units, if any.

Column	Example value	Type	Units
Platoon	3rd		
Ruck time	184.5		
Prior ruck marches completed	7		
Met the 12-mile standard	GO		

(c) In one sentence, state why the commander should require a data dictionary before anyone analyzes this roster.

3. Training and test sets. A cadet builds a model on all 64 company records, reports that it fits those 64 records very well, and recommends it to the commander. State what is wrong with that procedure and what the cadet should have done instead. Be specific about what the reported error is actually measuring. *5 pts*

4. Unusual or erroneous? Two records are flagged during cleaning. Classify each as an **unusual but legitimate** observation or a **data error**, and state what you would do with each. Use both statistical and contextual reasoning. *4 pts*

- **Record 19:** ruck time = 236 minutes
- **Record 41:** ruck time = 34 minutes

5. The modeling framework. For the commander's remedial-training problem, state the **decision**, the **objective**, and **one assumption** the model requires. *4 pts*

Part II — Descriptive Modeling

56 POINTS · APPROXIMATELY 32 MINUTES · LESSONS 2–7

Questions 6 through 9 use the eight recorded 12-mile ruck times, in minutes, from **1st Platoon**, listed in ascending order:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
168	172	175	175	180	184	190	236

6. Mean and median. Calculate the mean and the median. Show your work below the table. *6 pts*

Mean $\bar{x}$	Median

7. Sample standard deviation. Calculate the sample standard deviation s. Show the squared deviations. *8 pts*

$$s = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n - 1}}$$

s = _____ *(round to two decimal places)*

8. Quartiles, IQR, and range. Using the **median-of-halves** method (split the ordered data at the median; Q_1 is the median of the lower four values and Q_3 is the median of the upper four), complete the table. Show your work. *6 pts*

Q_1	Q_3	IQR	Range

Your range should be several times your IQR. In one to two sentences, explain what that gap tells you about this platoon, and state which of the two you would report to the commander as the spread of a typical cadet.

9. z-scores. Use **your own** $\bar{x}$ from Question 6 and s from Question 7. (If either is wrong you will receive follow-on credit here, so carry your values forward and continue.) *8 pts*

a. Calculate the z-score of $x_8 = 236$ and of $x_1 = 168$. Show your work.

z_8 = _____ z_1 = _____

b. Apply the course's **$|z| > 2.5$** screening rule. Which of these two observations, if either, is flagged as a potential outlier? Then explain what the flag does and does not establish.

10. Choosing a measure of center. The histogram below shows all 64 recorded ruck times in Company B. *6 pts*

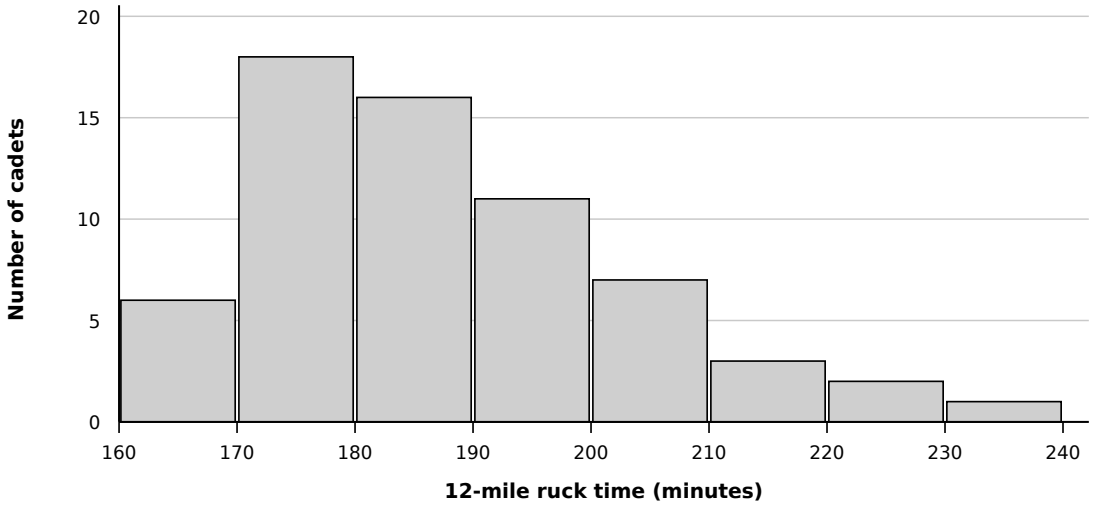


Figure 1 — Company B 12-mile ruck times, n = 64.

(a) Circle the measure of center you would report to the commander as the "typical" ruck time, and justify your choice in one to two sentences using the language of the course.

Mean **Median**

(b) Without computing anything, state whether the mean is larger or smaller than the median for this distribution, and why.

11. Boxplot versus histogram. Below is a boxplot of the eight 1st Platoon ruck times from Questions 6–9. *4 pts*

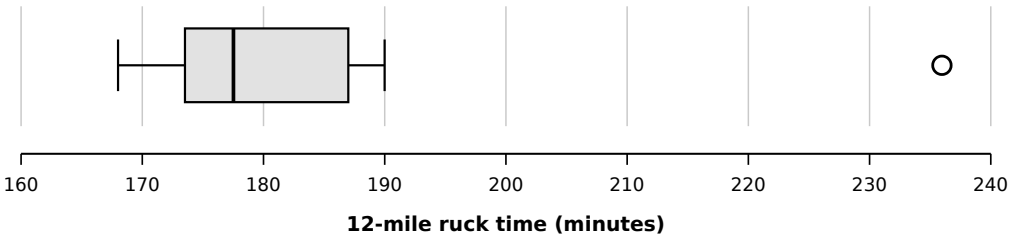


Figure 2 — 1st Platoon 12-mile ruck times, n = 8. The plotted point beyond the right whisker is flagged by the 1.5 × IQR fence.

(a) The box edges should match two values you computed in Question 8. Which two, and do they agree with the figure?

(b) What does the boxplot show that the histogram in Figure 1 does not, and what does the histogram show that the boxplot does not?

12. Reading a correlation table. Excel returns the correlations below, computed on all 64 records. Every cadet carried a 35-pound ruck; the load column varies only because of small differences in how canteens were filled. *8 pts*

Pearson correlation with 12-mile ruck time (minutes), n = 64

Variable	***r***
Weekly training mileage (mi/wk)	−0.92
2-mile run time (min)	0.78
Body weight (lb)	0.15
Ruck load (lb)	0.04

a. Which variable has the strongest linear association with ruck time? Interpret the **sign** in context, and state whether building a linear model on it is both **reasonable and necessary**.

b. A cadet concludes: “This proves that training more miles per week causes faster ruck times.” Is that conclusion supported? Name a plausible third variable.

c. Does $r = 0.04$ prove that ruck load has no effect on ruck time? Explain.

13. Write a caption. *10 pts*

You are preparing a one-page read-ahead for the battalion commander, who has no statistics background and must decide whether to fund a remedial ruck program. Write a caption for **Figure 1** (the histogram).

A figure caption should include a title and then, in three to five sentences, (1) describe the figure and (2) explain what the audience should take away from it — that is, why it matters for the decision. Your handwriting must be legible.

Part III — Linear Regression and Residuals

50 POINTS · APPROXIMATELY 32 MINUTES · LESSONS 9–16

Six cadets form the **training set**. For each, x is average weekly training mileage and y is the 12-mile ruck time.

Training set (n = 6)

Cadet	**1**	**2**	**3**	**4**	**5**	**6**
x — mileage (mi/wk)	5	10	15	20	25	30
y — ruck time (min)	210	195	189	183	177	180

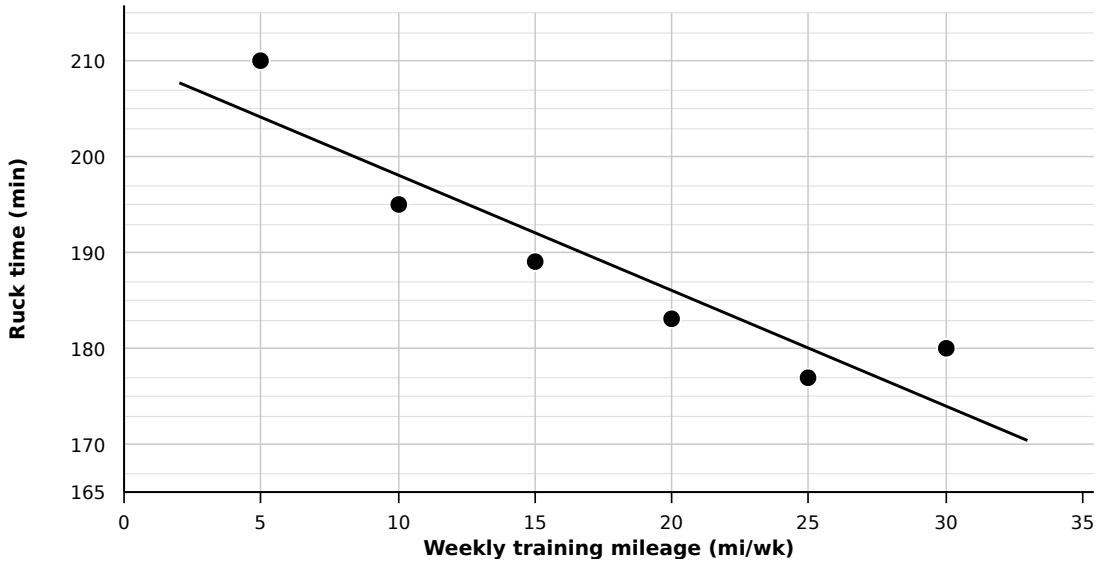


Figure 3 — Training set with least-squares trendline. Minor gridlines are 2 minutes.

14. Read the line of best fit off the graph. Do not use the data table for this question — read Figure 3. *6 pts*

(a) Read the value of the **trendline** (not the data points) at $x = 10$ and at $x = 30$.

at $x = 10$: $\hat{y} \approx$ _____ at $x = 30$: $\hat{y} \approx$ _____

(b) Use those two points to compute the slope, then write the equation of the line.

$\hat{y} =$ _____

Excel reports the trendline for the training data as $\hat{y} = 210 - 1.2x$, with $R^2 = 0.854$. Use these for the remainder of Part III.

15. Interpretation. *8 pts*

a. Interpret the **slope** in context, with units.

b. Interpret the **intercept** in context, and state whether it is meaningful here.

16. R^2. Complete the sentence, then state one thing R^2 does **not** tell you. *4 pts*

Approximately _____% of the variation in _____

is explained by its linear relationship with _____.

Would you describe this R^2 as strong, moderate, or weak? _____

17. Residuals and training error. *8 pts*

(a) Complete the four empty rows using $\hat{y} = 210 - 1.2x$.

Training residuals, $e = y - \hat{y}$

***x* (mi/wk)**	***y* actual**	**$\hat{y}$ predicted**	***e* residual**
5	210		
10	195	198	−3
15	189		
20	183	186	−3
25	177	180	−3
30	180		

(b) Compute the **training MAE** from all six residuals. Show your work.

Training MAE = _____ minutes

(c) Why does MAE use absolute values? What would happen if you simply averaged the six signed residuals?

18. Residual plot and the regression assumptions. *10 pts*

(a) Plot the six residuals from Question 17 against the **fitted values** $\hat{y}$ on the axes below.

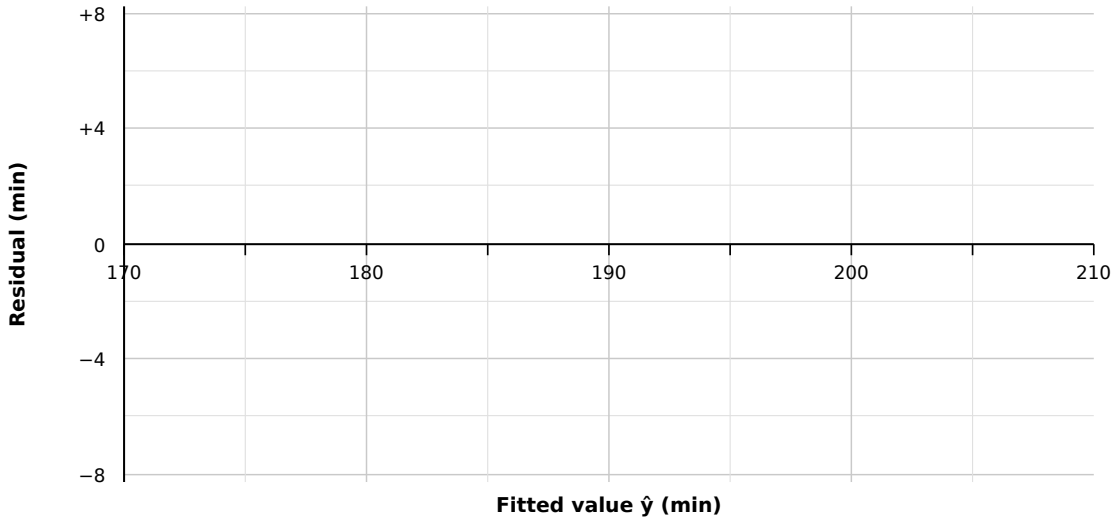


Figure 4 — Plot your residuals here. Minor gridlines are 5 minutes horizontally, 2 minutes vertically.

(b) Assess each assumption below, citing specific evidence from your plot.

Linearity — do the residuals scatter randomly around $e = 0$, or is there a systematic pattern?

Constant variance — is the vertical spread roughly equal across fitted values, or does it fan out or compress?

Outliers — are any residuals far outside the range of the others?

19. Distribution of residuals. Complete the table for the six training residuals, then answer below. *4 pts*

Statistic	Value (with units)
Mean of residuals	
Standard deviation of residuals	
Minimum residual	
Maximum residual	

The mean of the residuals should be very close to zero for a correctly fitted least-squares model. Is yours? What would it mean if the mean were substantially different from zero?

20. Test error versus training error. Three cadets were held out of the fit entirely. Complete the table using $\hat{y} = 210 - 1.2x$. *10 pts*

Test set (n = 3) — held out of the fit

x (mi/wk)	y actual	$\hat{y}$ predicted	$\lvert y - \hat{y} \rvert$
8	208		
18	183		
28	185		
Test MAE			

In two to three sentences, compare your test MAE against the **training MAE you computed in Question 17(b)** and state what the two numbers mean for using this model on this term's cadets.

Part IV — K-Nearest Neighbors

32 POINTS · APPROXIMATELY 22 MINUTES · LESSONS 19–21

The commander now wants to predict a simpler thing: will a cadet earn a **GO** on the 12-mile march? KNN requires a **categorical** dependent variable, so the continuous ruck time is converted to a class at the Army standard of 180 minutes.

KNN training set (n_{train} = 6)

Cadet	Weekly mileage (mi/wk)	2-mile run (min)	Ruck time (min)	Class
A	8	15	205	NO-GO
B	16	14	190	NO-GO
C	20	17	198	NO-GO
D	20	13	182	NO-GO
E	24	11	170	GO
F	32	14	176	GO

Cadet Q, whose class is unknown, trains **30 mi/wk** and runs 2 miles in **12.0 minutes**.

21. Set up the model. *6 pts*

a. The class label was created by splitting a continuous variable at a threshold. State the threshold used and why 180 minutes is a defensible choice here rather than, say, the median ruck time.

b. You will standardize both the training records and the test records. Explain why the test records must be standardized using the **training** mean and standard deviation, rather than statistics computed from the test set itself.

22. Standardize. *8 pts*

a. Compute the mean and sample standard deviation of **weekly mileage** across the six training cadets. Show your work.

$\bar{x}_{\text{mileage}}$ = _____ s_{mileage} = _____

b. For 2-mile run time you are given the training statistics $\bar{x}$ = 14.0 min and s = 2.0 min. Standardize **cadet Q** on both features.

z_1^* = _____ z_2^* = _____

23. Standardized distances. Complete the table, then show one full distance calculation below it. *8 pts*

Cadet	**z_1 mileage**	**z_2 run**	***d* from Q**	**Rank**	**Class**
A	−1.50	0.50			NO-GO
B					NO-GO
C					NO-GO
D					NO-GO
E					GO
F					GO

24. Choose k and classify. *6 pts*

(a) Apply the rule of thumb $k \approx \sqrt{n_{\text{train}}}$, rounded to the nearest **odd** integer.

$n_{\text{train}} =$ _____ $\sqrt{n_{\text{train}}} \approx$ _____ initial $k =$ _____

Why do we prefer an **odd** value of k for a two-class problem?

(b) Classify cadet Q by **majority vote** at each value of k. Name the neighbors used.

k	**Neighbors used (in order)**	**Vote**	**Predicted class**
1			
3			
5			

25. Sensitivity to k. Four held-out cadets were classified at each k. Their true classes are given; the accuracies are not. *4 pts*

Test cadet	**mi/wk**	**2-mi run**	**True class**	***k*=1**	***k*=3**	***k*=5**
Q	30	12.0	GO	*from Q24*	*from Q24*	*from Q24*
T2	12	17.0	NO-GO	NO-GO ✓	NO-GO ✓	NO-GO ✓
T3	28	13.0	GO	GO ✓	GO ✓	NO-GO ✗
T4	32	16.0	NO-GO	GO ✗	NO-GO ✓	NO-GO ✓
Test accuracy						

Carry your three predictions for cadet Q down from Question 24, mark them correct or incorrect, then fill in the three accuracies. Which k would you use, and why? Explain both what goes wrong at $k = 1$ and what goes wrong at $k = 5$.

Part V — Synthesis

8 POINTS · APPROXIMATELY 5 MINUTES · LESSONS 8, 16, 19–21

26. Compare all models. Complete the summary table. Write **N/A** where a metric does not apply to that model, and be prepared to say why it does not apply. *5 pts*

Model	R^2	Training MAE	Test MAE	Test accuracy
Baseline — predict the training mean (189 min)	N/A	9.00	9.67	
Linear regression, $\hat{y} = 210 - 1.2x$	0.854	*from Q17*	*from Q20*	
Degree-5 polynomial through all six training points	1.000	0.00	14.50	
KNN classification at your chosen k				*from Q25*

Which model do you recommend to the commander, and why? Address the degree-5 polynomial specifically.

27. Contrast the two model families. In two to three sentences, describe one fundamental difference between the way **linear regression** makes a prediction and the way **KNN** makes a prediction. What does each assume, or not assume, about the relationship between the inputs and the output? *3 pts*

END OF PRACTICE EXAMINATION

Check that you have answered all 27 questions. Turn the page for the continuation sheet.

CONTINUATION SHEET *— label each entry clearly with its problem number.*